The Centrality of Cultural Worldviews in the Climate Change Belief System Network

Introduction

Climate change remains one of the most politically polarized issues, particularly in the United States. Sharp divides exist among the public and policy elites regarding the existence of climate change, the extent of its risk, and what, if anything, should be done to address it. A consequence of political divides regarding climate change is stymied progress in reducing greenhouse gas emissions.

There are several explanations for why the public remains divided on climate change, including political ideology and partisanship; cultural worldviews derived from cultural theory; environmental orientation; acceptance/rejection of the scientific consensus on climate change; and an aversion to government regulations meant to address climate change. Each explanation involves beliefs about climate change, yet these beliefs do not exist in isolation from one another. Beliefs about climate change form an interconnected belief system.

A belief system is a set of interrelated and connected beliefs. One way to understand belief systems is to view them as existing within a hierarchy, in which higher-order beliefs influence or constrain lower-order beliefs (Nohrstedt et al. 2023; Peffley and Hurwitz 1985; Sabatier and Jenkins-Smith 1993). With the hierarchical approach to belief systems, the most abstract beliefs and general principles, termed *deep core beliefs* (e.g., political ideology), inform less abstract views about policy issues, *policy core beliefs* (e.g., climate change is happening), as well as attitudes about specific policy approaches, or *secondary beliefs* (e.g., support for a carbon tax). A second way of understanding belief systems is that of a network of connected beliefs (Boutyline and Vaisey 2017; Brandt, Sibley, and Osborne 2019; Brandt and Sleegers 2021; Fishman and Davis 2022). Similar to other types of networks, belief system networks have *nodes*, which are the beliefs, and *edges*, which are the connections, or relationships, between beliefs. Belief system networks can be analyzed using centrality measures to understand the importance and influence of the various beliefs (i.e., nodes).

Examining the public's climate change beliefs as a network can yield insights into polarized views. Indeed, several recent studies have examined the climate change belief system network (Lee et al. 2024, 2025; Lind, Morton, and Dalege 2024; Stewart, Rivera-Kientz, and Dacey 2024; Verschoor et al. 2020). However, apart from political ideology and partisanship, these studies did not include measures of other deep core beliefs such as cultural worldviews and environmental orientation, which have been shown to influence climate change beliefs (Jones 2011; Nowlin and Rabovsky 2020; Ziegler 2017).

In this paper, I use original survey data to examine the climate change belief system using measures of various types of beliefs not measured in previous research, including core beliefs (cultural worldviews, political ideology, partisanship, environment orientation), policy domain beliefs (views about climate change), and views on various policies to address climate change. By examining the climate change belief network, I can determine which beliefs are the most central within it. This allows comparison between the various leading explanations for the public's divergent views on climate. For example, if a recognition of the scientific consensus has high centrality in the network, that would indicate that the scientific consensus is a "gateway belief" that shapes other beliefs about climate change (Van Der Linden 2021).

Rather than adjudicating between competing hypotheses about which single type of belief – deep core, policy core, or secondary – should be most central, this paper takes an exploratory approach to mapping the structure of the climate change belief network. Prior theoretical traditions offer distinct expectations about where structural importance should concentrate. Hierarchical accounts point to deep core beliefs such as cultural worldviews and political identity. The gateway belief model points instead to perceived scientific consensus, and solution-aversion accounts point to policy preferences. I use these three traditions as theoretical anchors for interpreting which beliefs occupy structurally important positions in the network, rather than as mutually exclusive predictions to be confirmed or disconfirmed. This approach is likely better suited to revealing how different kinds of beliefs are organized relative to one another.

Using common measures of network centrality, including strength, closeness, and betweenness, I find that cultural worldviews, specifically egalitarianism (a deep core belief), support for EPA regulations (a secondary belief), and perceptions of climate change risk (a policy core belief) are the most central beliefs in the network. However, the centrality of beliefs varied across the different centrality measures, indicating that different types of beliefs serve different functions within a belief system network. For example, egalitarianism had the highest betweenness and closeness centrality among all respondents, indicating that it served as a bridge connecting other beliefs in the network. Yet, support for EPA regulations, a policy preference, had the highest strength centrality, meaning that it had the strongest connections (i.e., highest partial corrections with other nodes) in the network.

Belief Systems and Belief System Networks

Beliefs are propositions taken to be true, and the interconnections among them form a belief system. Early work on belief systems by Converse (1964) noted that belief systems consist of a "configuration of ideas and attitudes in which the elements are bound together by some form of constraint or functional interdependence" (3). Constraint or interdependence among idea-elements (i.e., beliefs) within a belief system means that one belief influences others, and a change in one belief is associated with a change in another. Converse (1964) also noted that beliefs can vary by centrality, where a belief that is "central" is less likely to change than a belief that is less central. More recently, Brandt and Sleegers (2021) argued that a theory of belief systems must contain three elements: a) the components of the belief system must be connected; b) these connections should be causal; and c) contain an accounting of the potential impacts of exogenous factors.

An early approach to understanding belief systems posited that belief systems have a hierarchical structure where higher-order beliefs, which are abstract beliefs and values, connect to and constrain lower-order beliefs about specific issues (e.g., Hurwitz and Peffley 1987, 1992; Peffley and Hurwitz 1985). Building on this work, policy scholars using the Advocacy Coalition Framework (ACF) have noted the importance of beliefs, structured hierarchically, for forming and maintaining advocacy coalitions, policy learning, and policy change (Nohrstedt et al. 2023; Sabatier and Jenkins-Smith 1993). Using ACF terminology, beliefs at the top of the hierarchy are known as deep core beliefs and can include political ideology, cultural worldviews, and environmental values, among others.

Deep core beliefs are foundational values that are sufficiently abstract that they can inform views on a variety of policy issues. As noted by Jenkins-Smith et al. (2014), core beliefs are:

> combinations of beliefs about broad societal norms concerning the role of government vs. the market in society, beliefs about the appropriate balance of expertise and centralized authority vs. communal or consensus-based decision making, and views about the relative priority of values such as liberty, equality, and security (492).

Additionally, deep core beliefs are central beliefs in the Converse (1964) sense and are, therefore, resistant to change. Finally, they act as filters for how information about policy issues is processed (Kahan 2012; Nowlin 2021).

While deep core beliefs have been measured in various ways, increasingly scholars are using cultural worldviews developed from cultural theory to conceptualize and measure them (Hornung and Bandelow 2021; Jenkins-Smith et al. 2014; Nohrstedt et al. 2023; Ripberger et al. 2014; Sotirov and Winkel 2016). In brief, cultural theory posits four cultural worldviews, or cultural types, which are derived from ideas about how societal relationships should ideally be structured. The four cultural types include *hierarchs*, who value clear lines of authority and rules that guide individual behavior; *egalitarians*, who value equality and collective well-being; *individualists*, who value autonomy and individual choice; and *fatalists*, who see the world as capricious and beyond human control. An individual's cultural worldview has been shown to inform their views on a host of specific policy issues, including national security (Ripberger, Jenkins-Smith, and Herron 2011); gun control (Kahan and Braman 2003); trade-offs between rights and security (Jenkins-Smith and Herron 2009); immigration (Jackson 2015); vaccines (Song 2014); and climate change (Jones 2011; Kahan et al. 2012; Nowlin and Rabovsky 2020).

Following deep core beliefs are *policy core beliefs* that involve beliefs about particular policy issues such as immigration, national security, or climate change. Policy core beliefs are influenced by deep core beliefs. Finally, *secondary beliefs* are beliefs regarding policy approaches such as a carbon tax or Environmental Protection Agency (EPA) regulations. Secondary beliefs are influenced by both deep core and policy core beliefs.

The hierarchical structure of beliefs includes the necessary elements of a theory of belief systems outlined by Brandt and Sleegers (2021), as the belief system components are connected and causality runs from higher-order (deep core) to lower-order (secondary) beliefs. Yet, one issue with the hierarchical structure conceptualization is that the boundaries between different types of beliefs are fuzzy. In addition, each type of belief has been measured in different ways, leading to different conclusions among scholars about belief systems (Henry et al. 2022; Ripberger et al. 2014). Conceptualizing belief systems as a network provides another way to understand their structure and organization that doesn't rely on hierarchical, vertical relationships among different types of beliefs. A network approach emphasizes the horizontal connections between different beliefs, and the influence of a belief on other beliefs is based on measures of centrality rather than their place in a hierarchy.

Belief System Networks

Within a belief system network, the components of the belief system – beliefs – are understood as *nodes*, and the connections between beliefs are *edges* or *ties*. Understanding belief systems as networks provides a way to operationalize belief systems consistent with how they are commonly defined and understood, and belief system networks can be analyzed using network centrality measures to determine which beliefs are prominent and influential in the network (Brandt and Sleegers 2021; Camina et al. 2022; Epskamp, Borsboom, and Fried 2018; Fishman and Davis 2022; Keskintürk 2022; Turner-Zwinkels and Brandt 2022; Verschoor et al. 2020).

A network approach can retain the notion of various "types" of beliefs (i.e., deep core, policy core, secondary) but shift the focus from their location in the hierarchy to their centrality in a belief system network. The centrality of the various types of beliefs highlights their importance and the function they perform in connecting other beliefs. Several studies have illustrated the usefulness of conceptualizing belief systems as networks, and many have found that "symbolic" beliefs (e.g., deep core beliefs such as political ideology) have high centrality values in belief system networks (Fishman and Davis 2022). For example, using the American National Election Study (ANES), Boutyline and Vaisey (2017) found that self-reported political ideology occupies a central role in a network of other beliefs about various policy issues (abortion, environment, guns, death penalty, immigration, inequality, welfare spending). Additionally, using common measures of network centrality, Brandt, Sibley, and Osborne (2019) find that symbolic beliefs based on identities (political ideology and partisanship) are more central within belief system networks than operational (issues-based) beliefs.

Beliefs that are more central in a belief system network keep an otherwise potentially disparate set of beliefs connected. Work by Boutyline and Vaisey (2017) posited a belief system network model where:

> (a) Individuals start with a single central belief ... (b) This central belief is used to produce a number of broad stances ..., which are then used to stochastically produce further beliefs. (c) Newly added sets of beliefs then form the basis for yet newer and more specific beliefs, repeating recursively to yield a center-periphery structure (1378-1379)

As the belief system forms a center-periphery structure, some beliefs can begin to act as bridges, connecting beliefs in the periphery to other beliefs in the network. Specifically, *betweenness* centrality measures how beliefs in the core of the network connect to those in the periphery. Multiple measures of centrality are typically used to determine the prominence of various beliefs in a belief system network. The most common measures of centrality, apart from betweenness, include *closeness* and *degree (strength)*. Each measure of centrality places distinct emphasis on what makes a node more or less important in the network, and that emphasis illustrates the function of the belief in the network.

Betweenness centrality measures the extent to which a node lies on paths between other nodes. A node high in betweenness is thought to be important because it reduces the distance between other nodes and, therefore, becomes an important link, or gatekeeper, to the other nodes (beliefs). The betweenness centrality equation for node n_i is

$$C_B(n_i) = \sum_{j<k} \frac{g_{jk}(n_i)}{g_{jk}}$$

where g_{jk} is the geodesic, or shortest path between nodes j and k, and $g_{jk}(n_i)$ is the number of shortest paths where n_i is present (Luke 2015; Quintana 2023). Beliefs that are high in betweenness centrality connect nodes in complex, center-periphery networks.

Closeness centrality measures how close nodes are in a network. Nodes that are closer to several other nodes are thought to be influential because the path from them to other nodes is short, which allows the node to more readily influence other nodes through a direct effect. The equation for the closeness centrality of node n_i is

$$C_C(n_i) = \left[\sum_{j=1}^{g} d(n_i, n_j)\right]^{-1}$$

where d is the distance between nodes. Closeness centrality is the inverse of the sum of distances between n_i and the other nodes in the network (Luke 2015; Quintana 2023). A larger value for closeness centrality indicates shorter distances to other nodes, which allows a belief to quickly influence other beliefs.

Degree centrality measures the number of direct connections to a node, with more connections being indicating the node's prominence and importance. Strength centrality is associated with degree centrality, but degree centrality is used in unweighted networks, or networks where connections are binary (i.e., yes or no), and strength centrality is used in weighted networks, where connections have a value, or weight, that denotes the strength of the connections. Specifically, strength centrality is calculated as the sum of the weights, rather than the sum of the connections, as in degree centrality. This means that nodes with high strength centrality are well-connected to other nodes.

In the following analysis, I examine the network of beliefs about climate change of a quota-based sample of the US public. Next, I briefly describe several drivers of the public's views on climate change.

Climate Change Beliefs

Even as climate scientists agree that climate change is happening, is anthropogenic, and poses significant risks, opinions are divided on each of those questions among some elites and some of the public. Several factors have been shown to influence beliefs about climate change among the public, including deep core beliefs such as partisanship, ideology, cultural worldviews, environmental attitudes, as well as policy core beliefs, including views about the scientific consensus, and secondary beliefs associated with aversions to the solutions presented to address climate change.

Perhaps the clearest divides regarding climate change are connected to political beliefs. Several studies have shown that, among the mass public in the US, conservatives and Republicans are less likely than liberals and Democrats to believe that climate change is happening or poses a risk (Bugden 2022; McCright and Dunlap 2011). Evidence suggests that polarization among the public is driven by elite polarization, as the public takes cues about policy and political issues from the political figures and news sources that they trust (Brulle, Carmichael, and Jenkins 2012; Carmichael and Brulle 2017; Krosnick et al. 2006; Merkley and Stecula 2018, 2021; Tesler 2018).

Apart from political beliefs, cultural worldviews derived from cultural theory have also been associated with climate beliefs. Early theorizing of cultural theory noted that the various cultural worldviews hold different "myths of nature", which shape how they view the natural environment and the risks posed to it (Thompson, Ellis, and Wildavsky 1990). In brief, egalitarians tend to view nature as fragile and ephemeral, whereas individualists see nature as robust. Hierarchs tend to see nature as tolerant within limits that need to be carefully and expertly managed. Finally, fatalists see nature as random and unpredictable and therefore impossible for humans to manage (Grendstad and Selle 2000). The various myths of nature associated with cultural worldviews likely drive climate beliefs. Research has found that egalitarians are the most concerned about climate change, and individualists are the least concerned among the cultural types (Jones 2011; Kahan et al. 2012; Kahan, Jenkins-Smith, and Braman 2011; Verweij et al. 2006). Hierarchs tend to occupy a "middle" space between egalitarians and individualists on environmental issues broadly (Ripberger et al. 2014) and climate change specifically (Nowlin and Rabovsky 2020). Finally, given their views on the capriciousness of nature, fatalists tend not to have clearly defined beliefs about climate change (Jones 2011; Nowlin and Rabovsky 2020).

Broader environmental values and attitudes also shape views on climate change. One of the most commonly used measures of environmental orientation is the New Ecological Paradigm (NEP) scale (Dunlap 2008; Sparks et al. 2021; Stern, Dietz, and Guagnano 1995). In brief, the NEP scale measures "beliefs about humanity's ability to upset the balance of nature, the existence of limits to growth for human societies, and humanity's right to rule over the rest of nature" (Dunlap et al. 2000, 427). Using the NEP scale, a pro-environmental orientation is associated with concerns about the robustness of nature and the impacts of human activities on the natural world. Previous research has found that a pro-environmental orientation, as measured by the NEP scale, is associated with views about climate change (Nowlin 2019; Ziegler 2017).

As noted, opinions differ on climate change despite the clear scientific consensus. A series of studies have found that acceptance of the scientific consensus around climate change acts as a "gateway belief" that connects to beliefs that climate change is happening and also connects to support for actions to address climate change (Goldberg et al. 2019; S. L. van der Linden et al. 2015; S. van der Linden, Leiserowitz, and Maibach 2019). Using experimental research designs, scholars have found that those exposed to a message about the scientific consensus were more likely than those not exposed to a consensus message to change their beliefs regarding climate change (i.e., it's happening, it's anthropogenic, and it's a risk) as well as increase their support for public action (Van Der Linden 2021).

Political beliefs are a leading source of division over climate change, and one explanation for their influence on both elites and the mass public is solution aversion (Campbell and Kay 2014). One of the underlying differences between Republicans/conservatives and Democrats/liberals is how they see the role of government, particularly the federal government, in addressing societal problems. Generally speaking, Republicans/conservatives are more skeptical of the government's ability to address problems than Democrats/liberals. Much of the discussion around climate change centers on government action to address it, and when government action is presented as a solution, it is likely to trigger skepticism among Republicans/conservatives, which in turn broadens skepticism about the problem. In essence, the solution aversion hypothesis posits that secondary beliefs and/or policy preferences drive broader views of policy issues. This is evidenced by the finding that the debate over the Kyoto Protocol polarized views about climate change (Krosnick, Holbrook, and Visser 2000).

Recent research using a network approach to examine climate change beliefs found that how worried respondents are about climate change, a policy core belief, was the most central belief in the climate change belief network (Lee et al. 2024). Additionally, support for regulating greenhouse gas emissions (secondary belief/policy preference) and beliefs about potential harm resulting from climate change (policy core belief) also scored high on measures of network centrality. Other studies found differences in the structure of climate change belief networks across groups, including those in global south who tended to have less density connected networks than those in the north (Lee et al. 2025), White, Black, and Hispanic respondents (Stewart, Rivera-Kientz, and Dacey 2024), Republicans and Democrats (Lee et al. 2024), and those on the political left vs the right in Denmark (Lind, Morton, and Dalege 2024). While these studies provide important insights into the climate change belief system of various publics, they did not include measures of many deep core beliefs shown to be important influences of climate change beliefs, including cultural worldviews and environmental orientation.

Theoretical Expectations

Previous research points to several distinct expectations about which types of beliefs should be most structurally central in a climate change belief network, though these expectations have not been tested against one another using measures of cultural worldviews, environmental orientation, and climate policy preferences within a single network. A hierarchical view of belief systems holds that deep core beliefs – cultural worldviews, political ideology, and pro-environmental orientation – shape and constrain other beliefs, and so should likely occupy the most central positions in the network. Work on the gateway belief model instead points to perceived scientific consensus, which is theorized to serve as a causal entry point that shifts downstream climate beliefs. A third possibility, grounded in solution aversion, is that beliefs about specific policy approaches are most central, because resistance to a policy solution can color how the underlying problem itself is perceived. Rather than adjudicating between these accounts as competing hypotheses, the analysis below examines centrality across all three types of beliefs simultaneously, asking which beliefs – and which types of beliefs – occupy the most structurally important positions in the network, and whether that varies depending on how centrality is measured. Betweenness and closeness centrality, which capture a belief's role in bridging and quickly reaching other beliefs in the network, most directly reflect the kind of constraining, foundational role a hierarchical account attributes to deep core beliefs. Strength centrality, which reflects the number and magnitude of a belief's direct connections, more directly reflects the pattern a solution-aversion account would predict for policy-specific beliefs.

In the next section, I discuss the data and measurements used to explore the climate change belief network.

Data and Analysis

To examine the belief system network around climate change, I use original public survey data collected in October 2017. The data comes from a representative sample, based on quotas for age, gender, and race/ethnicity drawn from the US Census, of about 1200 US residents obtained through Survey Sampling International (SSI). The survey was administered online through Qualtrics. Respondents were asked questions regarding their views on climate change, support for policy approaches to address climate change, agreement with statements from the New Ecological Paradigm (NEP), questions to determine their cultural worldview based on cultural theory, and their ideological and partisan attachments.

Although these data were collected in October 2017, several considerations support their continued relevance to current debates about the structure of climate change beliefs. As noted, deep core beliefs such as cultural worldviews are theorized to be especially resistant to change (Converse 1964), and the overall organization of the belief network – which beliefs are central and which are peripheral – should therefore likely be comparatively stable even as specific policy debates evolve. The network studies against which the present findings are compared also draw on data from a similar or earlier period (Verschoor et al. 2020). More directly, Lee et al. (2024) find that the structure of the climate change belief network is largely consistent over time, particularly among Republicans and Democrats considered separately. This suggests that partisan belief networks are not highly sensitive to the specific historical moment in which they are measured. Nonetheless, the 2017 data were collected early in the first Trump administration, prior to a number of subsequent developments in climate politics and communication, and it remains an open question whether more recent shifts in polarization and climate discourse have altered the network's structure. Future research should replicate this analysis with more recent data to establish whether the patterns reported here reflect an enduring feature of the climate change belief system or are specific to this period.

Belief Measures

Cultural theory, political ideology, partisanship, and the NEP scale were used to measure survey respondents' deep core beliefs. Questions regarding the belief and certainty that climate change is happening, the risk posed by climate change. and whether or not a scientific consensus exists about climate change were used to measure the policy core beliefs of respondents. Finally, policy preferences, including support for several policy approaches to address climate change, including a carbon tax, cap-and-trade, EPA regulations, international agreements, geoengineering, and nuclear energy, were used to measure secondary beliefs. I describe each of the measures below.[1]

For cultural worldview measures, I rely on a standard set of survey questions used in many previous studies (e.g., Jones 2011; Moyer and Song 2019; Nowlin and Rabovsky 2020; Tumlison and Song 2019), which have been demonstrated to have convergent, discriminant, and predictive validity (Swedlow et al. 2020). Specifically, respondents rated their agreement with 12 statements on a 1 (strongly disagree) to 7 (strongly agree) scale, presented in random order. Of the 12 statements, three represented each of the four cultural types. The statements for each cultural worldview are presented below.

[1] Summary statistics are shown in the appendix, Table A1.

The statements for the hierarchy worldview included:

- *The best way to get ahead in life is to work hard to do what you are told to do*
- *Society is in trouble because people do not obey those in authority*
- *Society would be much better off if we imposed strict and swift punishment on those who break the rules*

For egalitarianism, the statements included:

- *What society needs is a fairness revolution to make the distribution of goods more equal*
- *Society works best if power is shared equally*
- *It is our responsibility to reduce differences in income between the rich and the poor*

The individualism statements were:

- *Even if some people are at a disadvantage, it is best for society to let people succeed or fail on their own*
- *Even the disadvantaged should have to make their own way in the world*
- *We are all better off when we compete as individuals*

Finally, the fatalism statements included:

- *The most important things that take place in life happen by chance*
- *No matter how hard we try, the course of our lives is largely determined by forces beyond our control*
- *For the most part, succeeding in life is a matter of chance*

For analysis, I combined and averaged the three statements into a 1 to 7 scale for each worldview, with higher scores indicating greater agreement with the statements associated with that cultural type. The hierarchy scale has a mean (M) of 4.8 and a standard deviation (*sd*) of 1.4. In terms of reliability, the hierarchy scale has a Cronbach's α = 0.74. The egalitarianism scale has a M = 4.68, *sd* = 1.59, and a Cronbach's α = 0.82. For the individualism scale, the M = 4.48, *sd* = 1.44 and the Cronbach's α = 0.76. Finally, the fatalism scale has a M = 4.04, *sd* = 1.6, and a Cronbach's α = 0.8. Overall, the Cronbach's α values show sufficient reliability for each of the cultural worldview scales.

To measure political ideology, I asked respondents to self-identify on a 1 to 7 scale, with 1 = strongly liberal, 4 = middle of the road, and 7 = strongly conservative. The ideology scale has a M = 3.99, and a *sd* = 1.8. For partisanship, I asked respondents which party they most identified with, followed by how strongly they identified with that party. I combined the two questions into a 1 to 7 scale with 1 = strong Democrat, 4 = true independent, and 7 = strong Republican. Partisanship has a M = 3.88, and a *sd* = 1.96.

- **ideology**: *On a scale of political ideology, individuals can be arranged from strongly liberal to strongly conservative. Which of the following best describes your views?*
- **partisan**: *With which political party do you most identify? Do you completely, somewhat, or slightly identify with that party?*

To examine deep core beliefs related to the environment, I used three "pro-environment" statements from the NEP scale. As with the cultural worldview measures, respondents were asked to rate their agreement with the three statements on a 1 to 7 scale, and responses were averaged together into a single 1 to 7 scale. The combined scale has a M = 5.32, sd = 1.33, and a Cronbach's α = 0.75. The statements are listed below, and they were randomized on the survey.

- **New Ecological Paradigm Scale**:
  - *The balance of nature is very delicate and easily upset*
  - *Humans live on a planet with very limited room and resources*
  - *Humans are seriously abusing the environment*

For the policy core belief measures related to climate change, I examined whether respondents believe that climate change is happening (yes or no), and how sure they are that it is/is not happening (0 = Not at all sure, 1 = Somewhat sure, 2 = Very sure, 3 = Extremely sure). The questions, stated below, were combined into a nine-point scale with -4 indicating that respondents are *extremely sure* that climate change is *not* happening to 4 indicating that respondents are *extremely sure* that climate change *is* happening. The *happening* measure has a M= 2.37, and a sd = 2.1.

In addition, respondents were asked about their perception of risks associated with climate change on a 0 (no risk) to 10 (extreme risk) scale, and the *risk* measure has a M = 6.95 and a sd = 2.76. Finally, I asked respondents their perceptions of the scientific consensus surrounding climate change. The *sciconsensus* measure is dichotomous and has a M = 0.63 (sd = 0.48) indicating that 63 percent of respondents believe there is a scientific consensus on climate change. Exact question wordings and variable labels are below.

- **happening**: *Do you think that climate change is happening? How sure are you that climate change [is/is not] happening?*
- **risk**: *On the scale from zero to ten, where 0 means no risk and 10 means extreme risk, how much risk do you think climate change poses for people and the environment?*
- **sciconsensus**: *Most scientists think climate change is happening*[2]

It is worth noting that this operationalization of perceived scientific consensus differs from the framing used in much of the gateway belief model literature, which typically presents respondents with an explicit numerical consensus estimate (e.g., that 97 percent of climate scientists agree that climate change is happening) before asking about downstream beliefs (S. van der Linden, Leiserowitz, and Maibach 2019). The sciconsensus measure used here instead asks respondents for their own naturalistic belief about whether such a consensus exists, without providing a specific consensus figure. This is a more conservative measure in the sense that it does not prime respondents with the consensus estimate itself, so it may better capture organically held beliefs about scientific consensus rather than beliefs induced by exposure to a consensus statistic.

Finally, for secondary beliefs/policy preferences, I measured support for various policy approaches using a 1, strongly oppose to 7, strongly support scale. Policies were presented in random order.

[2]The exact wording for the scientific consensus question was: Which of the following statements comes closest to your view? 1 – Most scientists think climate change is happening, 2 – There is disagreement among scientists, 3 – Most scientists think climate change is not happening, 4 – Don't know. I then created the `sciconsensus` dummy variable with 1 = Most scientists think climate change is happening, and 0 = all other responses.

Various policy solutions have been proposed as ways for the federal government in the United States to deal with climate change. Using a one to seven scale, where 1 is strongly oppose and 7 is strongly support, please indicate your support for each of the following policy proposals.

- **IntAgree**: *Accept internationally-established limits on U.S. production of carbon dioxide and other greenhouse gases thought to cause climate change*
- **Renew**: *Encourage research and development of renewable energy sources like solar or wind power*
- **Nuclear**: *Expand the use of nuclear energy*
- **Tax**: *A carbon tax that imposes a charge on coal, oil, and natural gas in proportion to the carbon they contain*
- **EPA**: *Environmental Protection Agency regulations that limit the amount of carbon dioxide and other greenhouse gases that power plants can emit*
- **CapTrade**: *A cap and trade program that sets an overall cap on emissions through allowances that can be bought, sold, or saved for future use*
- **GeoEng**: *Encourage development of "geoengineering" technology such as filters that remove excess carbon dioxide from the air and high-altitude orbiting reflectors to reduce solar heating*

Overall, renewable energy has the highest mean at 5.89 (*sd* = 1.39). The next highest level of support is for US Environmental Protection Agency (EPA) regulations at M = 5.35 (*sd* = 1.66), followed by geoengineering at M = 5.32 (*sd* = 1.55), and support for an international agreement at M = 5.07 (*sd* = 1.73). For the two market-based approaches, support for a carbon tax has a M = 4.79 (*sd* = 1.84), and support for cap-and-trade is M = 4.75 (*sd* = 1.71). Finally, support for expanding nuclear energy has the lowest level of support at M = 4.09 (*sd* = 1.92).

Belief System Network Analysis

Examining belief systems as networks involves the same types of assumptions and analysis as examining other types of networks, such as social or informational networks. A network is composed of nodes (e.g., groups, individuals, information, beliefs) and edges, or ties that connect the various nodes. Additionally, networks can be directed, where the connections are only in one direction ($\mathbf{A} \rightarrow \mathbf{B}$), or undirected with reciprocal relationships ($\mathbf{A} \leftrightarrow \mathbf{B}$). Networks can be unweighted, where edges take on a binary value (i.e., a connection is present or not) or weighted, where the edges are assigned a value. In a belief system network, nodes represent observed variables, and the edges represent statistical relationships between them (Epskamp, Borsboom, and Fried 2018). In addition, they are often weighted networks, and the values assigned to the edges are based on the strength of the relationship between the nodes. Finally, belief system networks are typically exploratory and therefore often assumed to be undirected (Costantini et al. 2015).

In the following analysis, I estimate the climate change belief system network using a Gaussian graphical model (GGM), a type of pairwise Markov random field (PMRF) model commonly used to examine psychological networks (Epskamp 2017). In a GGM, the edges of the network represent partial correlation coefficients that estimate the association between nodes while controlling for the other nodes in the network. The input for the GGM is a covariance matrix, but the network model is based on the inverse of the covariance matrix, which produces the partial correlation estimates (Epskamp and Fried 2018). Polychoric correlation is used when the data contain ordinal variables. The model is a weighted network, and it is weighted by the strength of the partial correlations. It is then regularized using the "least absolute shrinkage and selection operator" (LASSO), which reduces the possibility of spurious correlations by reducing small coefficients to essentially zero (Brandt, Sibley, and Osborne 2019; Friedman, Hastie, and Tibshirani 2008). This approach is common in psychological network analysis because of the small sample size relative to the data needed to estimate network models (Epskamp and Fried 2018). The LASSO requires a tuning parameter for regularization to manage the sparsity of the network. A common approach is to use an information criteria to select the appropriate parameter, such as the extended Bayesian Information Criterion (EBIC), which provides additional penalties for model complexity (Epskamp 2017; Foygel and Drton 2010). To estimate and illustrate the network, I use the `bootnet` and `qgraph` packages in `R` (Epskamp et al. 2012; Epskamp and Fried 2023).

The measures of centrality – betweenness, closeness, and strength – in the belief system network were calculated through a bootstrapping procedure using 1000 resamples of the data. The procedure was performed in `R` using the `bootnet` package (See Epskamp and Fried 2023, 2018). The bootstrap method results show the estimated centrality and the 95% confidence interval for each estimate.

The GGM is an undirected model. It estimates whether two beliefs are conditionally related but not the direction of that relationship. This is the standard approach in exploratory belief-network research, where strong a priori commitments about which specific beliefs causally precede others are typically unavailable (Epskamp, Borsboom, and Fried 2018). However, because a central question in this paper is whether beliefs are better understood as organized hierarchically or as a network, it is useful to assess whether the network's structure – and in particular, whether a belief is estimated to be highly central – depends on this choice of an undirected model. I therefore estimate a companion Bayesian network (BN) using the same 17 belief nodes.[3] Unlike the GGM, a BN is a directed acyclic graph, in which edges represent an estimated ordering between beliefs. This offers a way to assess whether a given belief occupies an upstream, "parent" position relative to others – the pattern a hierarchical account of belief systems would predict for deep core beliefs in particular. I learn the network structure using hill-climbing, tabu search, and constraint-based (PC) algorithms, each optimized via BIC or conditional independence testing, and assess stability with a nonparametric bootstrap (500 resamples), retaining edges and their estimated direction that appear in at least 50% of resamples to construct an averaged network. Robustness to missing data is checked via multiple imputation ($m = 5$). For each node, I calculate its out-degree (the number of other beliefs relative to which it occupies an upstream, "parent" position in the estimated graph) and the size of its Markov blanket (the parents, children, and co-parents needed to fully predict it) as BN analogues to the GGM's centrality measures.

A caveat on directionality. These algorithms are applied to cross-sectional, purely observational survey data. There is no temporal ordering, experimental manipulation, or causal-identification assumption (e.g., non-Gaussian error structure, as in LiNGAM) built into the analysis, so the estimated edge directions are not causally identified. Score-based algorithms such as hill-climbing and tabu search return a fully directed acyclic graph even when several directed graphs fit the data equally well (i.e., are Markov equivalent). The direction chosen is whichever fits the search criterion marginally better, not one the data have distinguished from its reverse. Constraint-based algorithms such as PC can identify some edge directions unambiguously

[3]Bayesian network structure learning was conducted using the `bnlearn` package in `R` (Scutari 2010). Continuous variables were discretized into tertiles (or, for `happening` and `risk`, substantively meaningful categories) for the score-based algorithms; a Gaussian approximation avoiding discretization was run as a sensitivity check.

but generally leave others unresolved absent further assumptions. “Upstream” and “downstream” positions below should therefore be read as a property of how a belief’s distribution relates to others’ under the model’s search criteria, not as evidence that one belief causes another.

In the following analysis, I first examine the climate change belief network, which includes deep core beliefs (cultural worldviews, ideology, partisanship, NEP scale), policy core beliefs (climate change is happening, climate change risk, scientific consensus), and secondary beliefs/policy preference (support for various policy measures). Then, I explore the betweenness, closeness, and strength centrality measures of the nodes in the climate change belief network to identify which beliefs are most central.

Data and Code Availability

This study was not preregistered. No a priori power analysis was conducted; the bootstrap correlation-stability coefficients reported in the Results section, all at or above the 0.50 threshold Epskamp et al. (2018) recommend, serve as a post hoc check that the sample size supports stable centrality estimates. The data and analysis code are organized in a version-controlled repository – including the raw and derived survey data, all analysis scripts, and the manuscript source – with package versions pinned via `renv` to support exact reproducibility.

Results

The results for the climate change belief network are shown in Figure 1. The results depicted in Figure 1 show deep core beliefs in black, policy core beliefs in gray, and secondary beliefs/policy preferences in white. Additionally, positive partial correlations between nodes are represented by a solid line, and negative partial correlations are represented by a dashed line. Finally, the network is weighted, and the strength of the partial correlations is illustrated by the thickness of the edges, with thicker paths illustrating stronger correlations.

Overall, the network has 17 nodes and 85 (of 136) non-zero edges, or connections. Looking at the network overall, it seems that the cultural worldview measures are clustered together, the political belief measures are also clustered, the NEP scale is clustered with the policy core measures (i.e., climate change beliefs), and the secondary beliefs/policy preferences (i.e., climate policy measures) are clustered together, with the exception of nuclear energy. The directions of the relationships in the network are as expected, with deep core beliefs associated with climate beliefs, and both are associated with support for various climate policies. Finally, some of the stronger associations in the network are between partisanship (more Republican) and ideology (more conservative); beliefs that climate change is happening is strongly associated with beliefs about the scientific consensus as well as beliefs about the risks of climate change; the various policy measures (aside from nuclear energy) are also highly correlated, particularly support for renewable energy and geoengineering; and the cultural worldviews of hierarchy and individualism as well as egalitarian and fatalism are highly correlated. After visualizing the network, I next examine various centrality measures within it. The results of the bootstrap procedure for estimating the centrality measures and the 95% confidence intervals are shown in Figure 2.

Looking first at betweenness centrality, as shown in Figure 2, egalitarianism is the node with the highest betweenness centrality, by a wide margin, with 61.89. Next is climate change risk perceptions at 41.19, followed

Figure 1: Belief System Network

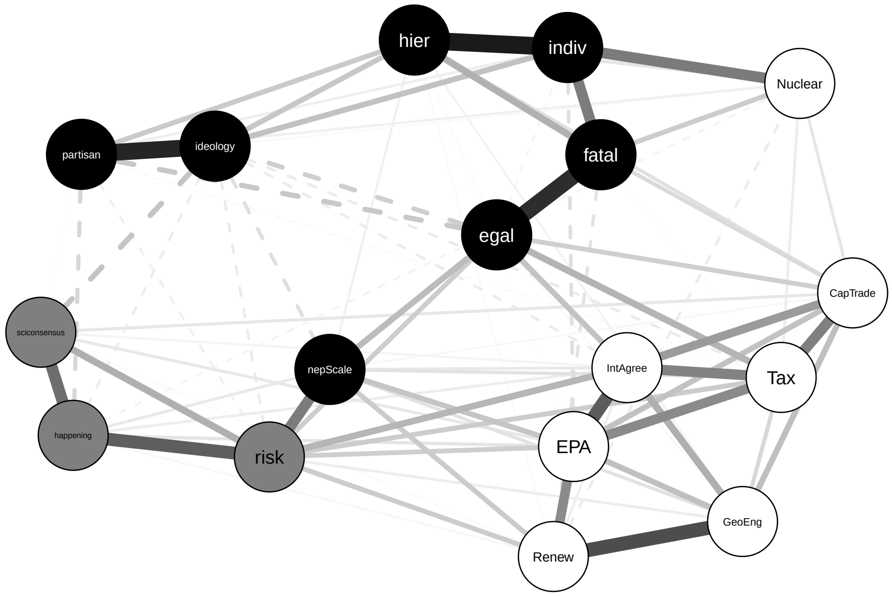
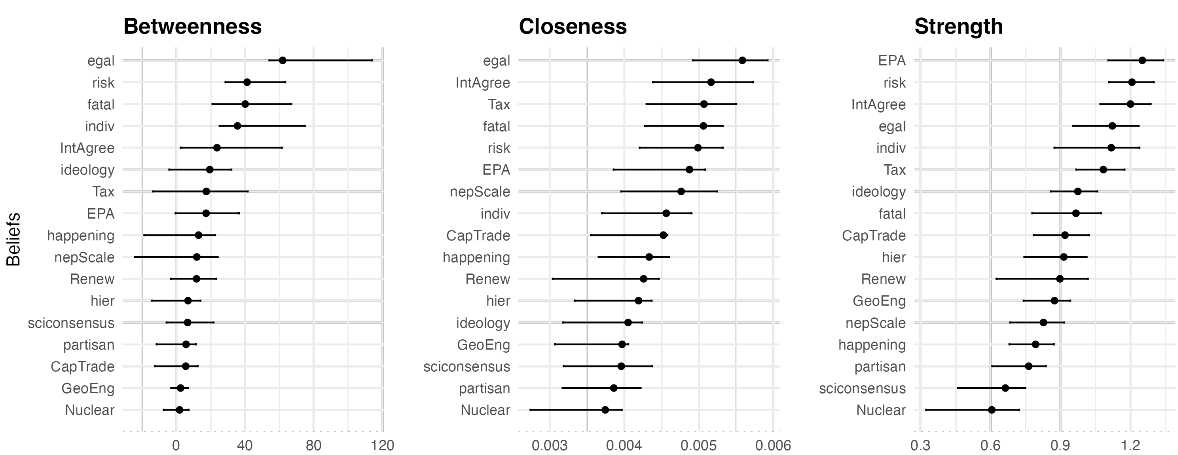


Figure 2: Measures of Network Centrality



closely by fatalism at 40.11, and individualism with 35.62. Nodes high in betweenness sit on the shortest path between the other nodes in the network and act as a bridge between other beliefs; therefore, egalitarianism works to connect beliefs in the network.

Also shown in Figure 2, egalitarianism has the highest closeness centrality with 0.00558, followed by support for an international agreement to address climate change at 0.00516. Beliefs high in closeness are more closely connected to other nodes, so they have the shortest paths to them, which allows them to be influential in the network.

Finally, looking to strength centrality, support for EPA regulations has the highest degree of strength centrality, followed by perceptions of the risk posed by climate change, and support for an international agreement to address climate change. Strength centrality reflects the sum, using absolute values, of weighted connections. The EPA node has the highest degree of strength centrality among the nodes in the network, with 1.25. Of note, the highest partial correlation coefficients for the EPA node included the policy support variables international agreement at $r = 0.28$, carbon tax at $r = 0.201$, and renewable energy at $r = 0.201$, as well as the NEP scale at $r = 0.099$. The support for an international agreement had a strength centrality of 1.21 and, similar to EPA regulations its strongest partial correlations were with the other policy support measures including EPA regulations (0.28), a carbon tax (0.216), cap-and-trade (0.172), and geoengineering (0.138) as well as policy domain belief of climate change risk (0.127) and the core belief of egalitarianism (0.104). Finally, the climate change risk node had a strength centrality of 1.2. It was most highly correlated with the other policy core beliefs, including the belief that climate change is happening and that there is a scientific consensus on climate change.

GGM Robustness Checks

To assess the stability of the reported network and its centrality estimates, I conducted four additional checks. First, a case-dropping bootstrap procedure estimated the correlation stability (CS) of each centrality measure – the proportion of cases that can be dropped while maintaining a correlation of at least 0.7 with the full-sample estimates. The CS coefficients for betweenness (0.52), closeness (0.75), and strength (0.75) all meet or exceed the 0.50 threshold Epskamp et al. (2018) consider stable.

Second, I re-estimated the network across a range of EBIC tuning values (gamma = 0, 0.25, 0.5, 0.75, 1.0) to check whether the reported results depend on the tuning = 0.5 default used above. The number of non-zero edges ranges from 79 to 105 across this range, but the node with the highest strength, betweenness, and closeness centrality is unchanged at every value tested.

Third, I compared the EBIC-selected network to one selected via cross-validation, using the `huge` R package. The cross-validated network has 86 edges, compared to 85 from the EBIC-selected network – a materially similar level of sparsity.

Finally, I conducted pairwise bootstrap difference tests for all 136 possible node pairs on both betweenness and strength (See Epskamp and Fried 2023). 31 of 136 pairwise comparisons were statistically significant for betweenness, and 82 of 136 were significant for strength. Among the five nodes with the highest betweenness centrality, none of the 10 pairwise comparisons reached significance, meaning their betweenness centrality is not statistically distinguishable from one another. Among the five nodes with the highest strength centrality, only 1 of 10 pairwise comparisons was significant.

Bayesian Network Robustness Check

To assess whether these conclusions depend on the choice of an undirected GGM, I estimated a companion Bayesian network using the same 17 belief nodes (see Methods). Figure 3 shows the averaged, bootstrap-stable directed network (threshold = 0.50, 23 arcs retained, drawn from 27 hill-climbing arcs, 28 tabu-search arcs, and 26 PC-algorithm arcs). 8 arcs were recovered by all three algorithms, and 20 arcs were stable across all five multiply-imputed datasets.

Figure 3: Averaged Bayesian Network (bootstrap threshold = 0.50)

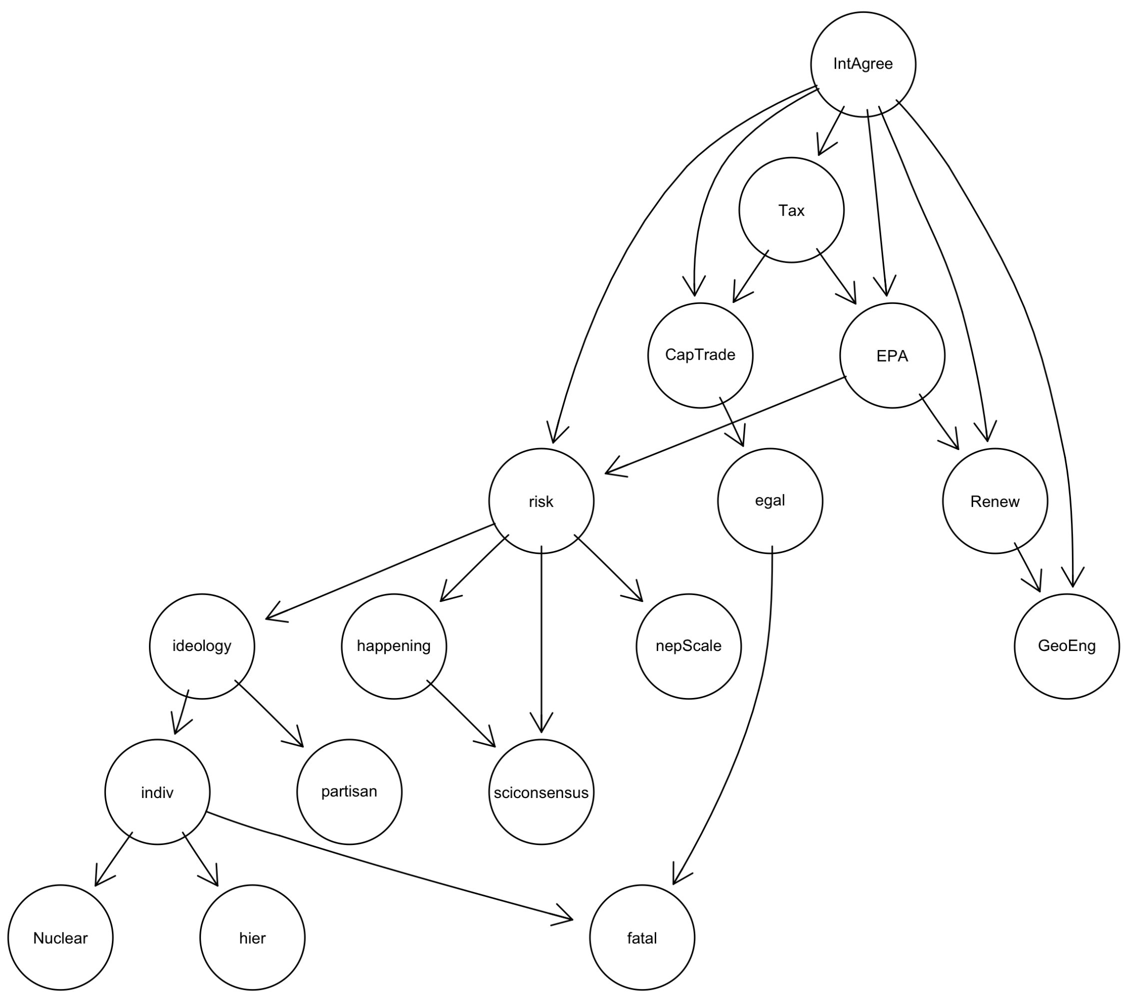


Table 1 reports, for each node, its position in the directed network (in-degree, out-degree, and Markov blanket size) alongside its rank on each GGM centrality measure (1 = most central). The comparison reveals both convergence and divergence between the two approaches. Support for an international agreement (`IntAgree`), a secondary/policy belief, has the highest out-degree in the network. It occupies an upstream, "parent" position relative to more other beliefs than any other node, including several deep core cultural worldviews and policy preferences (see the caveat on directionality above: this describes its estimated position in the graph, not a claim that it causes those other beliefs). Individualism (`indiv`) and perceived climate risk (`risk`) also rank highly on both GGM centrality and BN out-degree/Markov blanket size, indicating that their structural importance is likely not an artifact of the undirected model. Egalitarianism, by contrast, had the highest betweenness and closeness centrality in the GGM, yet it ranks well outside the top nodes on every BN measure. It appears more as a well-connected but largely downstream node than as an upstream, "parent" belief. I return to the implications of this divergence in the Discussion.

Table 1: GGM Centrality Ranks vs. Bayesian Network Structural Position

Node	Type	BN In-Degree	BN Out-Degree	BN Markov Blanket	GGM Betweenness Rank	GGM Closeness Rank	GGM Strength Rank
IntAgree	Secondary	0	6	6	5	2	3
risk	Policy Core	2	4	6	3	5	2
indiv	Deep Core	1	3	5	2	8	6
EPA	Secondary	2	2	4	6	7	1
ideology	Deep Core	1	2	3	7	14	7
Tax	Secondary	1	2	3	7	3	5
Renew	Secondary	2	1	3	9	13	12
CapTrade	Secondary	2	1	3	13	10	9
egal	Deep Core	1	1	3	1	1	4
happening	Policy Core	1	1	2	11	9	14
fatal	Deep Core	2	0	2	4	4	8
sciconsensus	Policy Core	2	0	2	10	12	16
GeoEng	Secondary	2	0	2	11	16	11

Table 1: GGM Centrality Ranks vs. Bayesian Network Structural Position

Node	Type	BN In-Degree	BN Out-Degree	BN Markov Blanket	GGM Betweenness Rank	GGM Closeness Rank	GGM Strength Rank
hier	Deep Core	1	0	1	13	11	10
nepScale	Deep Core	1	0	1	13	6	13
partisan	Deep Core	1	0	1	13	15	15
Nuclear	Secondary	1	0	1	13	17	17

Discussion and Conclusion

Climate change is a polarized issue among the US public as beliefs about climate change diverge among the public. Beliefs about climate change form an interconnected belief system. One way to understand belief systems is through a hierarchical approach in which foundational, abstract beliefs influence other beliefs about specific issues and policy approaches. Another way to understand the structure of belief systems is as a network, with beliefs as nodes and connections between beliefs as edges.

In this paper, I used original survey data from a quota-based sample of the US public to examine the belief system network associated with climate change. Additionally, I drew on the leading influences of climate change views identified in the literature, including political beliefs, cultural worldviews, pro-environment orientation, acceptance of the scientific consensus, and solution aversion, as theoretical anchors for exploring which beliefs would be the most central in the network using the common measures of centrality, including betweenness, closeness, and strength.

Rather than treating these patterns as confirming or disconfirming specific hypotheses, they are better read as evidence about where structural importance concentrates in the climate change belief network. Cultural worldviews, policy preferences, and perceived climate risk each occupy central positions on at least one measure of centrality, while perceived scientific consensus does not stand out on any of them. This pattern is broadly consistent with theoretical accounts that emphasize deep core beliefs and policy-specific resistance as organizing features of belief systems, and less consistent with an account in which scientific consensus functions as the primary structural hub. At the same time, the pairwise comparisons reported in the Results section show that many of these differences are not statistically distinguishable from one another. Egalitarianism's betweenness centrality, for example, is not significantly different from that of individualism, fatalism, perceived risk, or support for an international agreement. Structural importance in this network is therefore likely concentrated among a cluster of beliefs spanning multiple theoretical categories, rather than residing in any single belief or type of belief, and claims about which specific belief is "most central" should be made cautiously.

As noted, this null finding regarding perceived scientific consensus deserves a more careful interpretation. The gateway belief model is a causal, interventional framework. It is typically tested using experimental or longitudinal designs that trace how exposure to a consensus message shifts later beliefs (Said, Frauhammer, and Huff 2022). The present analysis instead estimates a cross-sectional, undirected network, and centrality in this type of network reflects structural embeddedness, not causal influence. A belief can be a causal entry point for belief change without also being central to how beliefs are organized at a given point in time – these are different properties. Recent work also shows that the gateway belief model's effects are conditional rather than universal, likely varying with respondents' metacognitive engagement with the consensus message (Said, Frauhammer, and Huff 2023) and with the specific mechanisms and boundary conditions of the messaging itself (Göbel et al. 2026). The low centrality of scientific consensus in the present network, then, does not contradict the gateway belief model. It is likely, instead, that consensus beliefs are not structurally embedded at the core of the broader climate belief system, even though they can still function as effective causal levers under targeted informational interventions. Future research should combine network and experimental approaches to adjudicate between the structural and causal accounts of the gateway belief model.

For the measures of closeness and betweenness centrality, egalitarianism was the most central node in the network. Closeness and betweenness centrality both address the lengths of the paths between nodes, with closeness meaning that a node influences other nodes because of its short path, and betweenness meaning that a node is influential because it sits on the path between other nodes. Beliefs high in closeness and betweenness act as gateways to other beliefs. These findings are consistent with the notion that cultural worldviews are deep core beliefs structurally connected to other beliefs. Egalitarianism in particular has been shown to be associated with environmental and climate change views (Ellis and Thompson 1997; Grendstad and Selle 1997; Jones 2011; Nowlin 2022; Nowlin and Rabovsky 2020; Xue et al. 2014). Additionally, the betweenness centrality finding of egalitarianism is also consistent with cultural cognition, in which cultural worldviews shape not only *what* people think but also *how* they think and process information about political and policy concerns (Kahan 2012; Kahan et al. 2012; Kahan, Jenkins-Smith, and Braman 2011).

Strength centrality is calculated as the sum of the absolute values of the weighted edges, and the sum reflects either the node with the strongest correlations and/or the node with the highest number of edges. As shown in Figure 2, the EPA node has the highest strength centrality, followed closely by the climate change risk node, and finally the international agreement node. The strength centrality of views about EPA regulations and an international agreement to address climate change support the importance of solution aversion – opposition to climate policies – within the climate change belief system.

Finally, the results showed that the centrality of the various beliefs differed across the centrality measures. As noted, egalitarianism had a higher level of betweenness and closeness centrality, whereas EPA regulations had the highest level of strength centrality. This suggests that different types of beliefs likely serve distinct functions within a belief system. The functions of a belief in a network may be multifaceted and/or vary across policy issues, meaning that a hierarchical approach may be appropriate for some issues and contexts but not for others. For example, in the climate change belief network examined here, egalitarianism, a deep core belief, acted as a bridge, connecting beliefs in the network, as indicated by the betweenness centrality measure. This finding is consistent with a hierarchical approach to belief systems, where deep core beliefs act as filters and constrain other beliefs. However, the strength centrality of EPA regulations is not, as a hierarchical approach would posit that such secondary beliefs would be on the periphery of the network, and deep core or policy core beliefs would be central. These results point to both deep core beliefs and policy beliefs occupying structurally important positions in the belief system, which would not be predicted by a hierarchical approach.

The Bayesian network robustness check complicates this picture further, and in a way that is instructive. Egalitarianism's betweenness centrality in the undirected GGM was read above as consistent with a hierarchical account, in which a deep core belief acts as a bridge that connects and constrains other beliefs in the network. That reading does not survive a model capable of estimating direction. In the BN, egalitarianism has neither a notably high out-degree nor a large Markov blanket, meaning it does not occupy an upstream, "parent" position in the estimated graph relative to the rest of the network. This likely reflects egalitarianism's structural position – bridging the cluster of cultural worldview items to the rest of the network – rather than the kind of directional, constraining role a hierarchical account would attribute to a deep core belief. The stronger challenge to a hierarchical reading comes instead from `IntAgree`. Support for an international climate agreement, a secondary/policy-level belief, has the highest out-degree of any node in the directed network, meaning it occupies a more upstream position in the estimated graph than every deep core and policy core belief in the model, including egalitarianism (a statement about estimated graph position under the algorithms described in the Methods, not a demonstrated causal claim – see the caveat on directionality there). A hierarchical account offers no obvious explanation for why a specific climate policy preference would occupy this position. At the same time, individualism and perceived climate risk are prominent by both approaches, so the evidence does not simply favor a network account over a hierarchical one. Whether a belief's position is consistent with a hierarchical account likely depends on the specific belief in question, and centrality in an undirected network is not, by itself, reliable evidence of hierarchical position. Future research should use panel data or explicit causal-identification designs to test whether these estimated positions reflect actual causal precedence, and to clarify when a directed or undirected framework is more appropriate.

While these results shed light on the structural importance of various beliefs within the climate change belief network, several avenues for future research remain. Of note is how the various climate policy measures are highly correlated, with the exception of nuclear energy, which was only weakly positively correlated with support for cap-and-trade and a carbon tax, and weakly negatively correlated with support for renewable energy. However, support for nuclear energy was strongly positively correlated with individualism and weakly positively correlated with fatalism. This follows other research that has shown that views about nuclear energy are not associated with concern about climate change, even though nuclear energy is a low-carbon energy source (Feldman and Hart 2018; Hawes and Nowlin 2022; Ter-Mkrtchyan et al. 2022).

The results shown here illustrate the centrality of beliefs across the entire sample; however, it is possible that some subgroups have distinct belief networks, with different beliefs being more (or less) central than in the aggregate. For example, Camina et al. (2022) found that the structure of belief system networks varied across different subgroups, including liberals and conservatives, younger and older, and male and female. Looking specifically at climate change, Lee et al. (2024) compared the belief system networks of Democrats and Republicans and found that they had a similar structure and density compared to independents and those with no party affiliation. Additionally, Stewart, Rivera-Kientz, and Dacey (2024) found differences in network structure and belief centrality across racial groups, including Whites, Blacks, and Hispanics. Additional research with a larger sample size should examine possible differences in networks across the various cultural types.

These findings speak to debates in both policy process research and climate communication. For policy process scholars, the results suggest that the choice between hierarchical and network conceptions of belief systems may depend on the specific beliefs under study rather than being a general feature of belief systems as a whole. This complicates any single framework's claim to universal applicability and points toward using both undirected and directed network methods together to characterize belief system structure. For climate communication research, the finding that perceived scientific consensus is not structurally central, together with the reframing of the gateway belief model discussed above, suggests that consensus messaging campaigns may be better understood as targeted causal interventions than as efforts to shift a structurally central belief – a distinction with direct implications for how such campaigns are designed and evaluated.

Appendix

The descriptive statistics for each of the variables are presented in Table A1.

Table A1: Descriptive Statistics for Belief Measures

	Mean	SD	Min	Max
hier	4.80	1.40	1.00	7.00
egal	4.68	1.59	1.00	7.00
indiv	4.48	1.44	1.00	7.00
fatal	4.04	1.60	1.00	7.00
ideology	3.99	1.80	1.00	7.00
partisan	3.88	1.96	1.00	7.00
nepScale	5.32	1.33	1.00	7.00
happening	2.37	2.10	−4.00	4.00
risk	6.95	2.76	0.00	10.00
sciconsensus	0.63	0.48	0.00	1.00
IntAgree	5.07	1.73	1.00	7.00
Renew	5.89	1.39	1.00	7.00
Nuclear	4.09	1.92	1.00	7.00
Tax	4.79	1.84	1.00	7.00
EPA	5.35	1.66	1.00	7.00
CapTrade	4.75	1.71	1.00	7.00
GeoEng	5.32	1.55	1.00	7.00

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